

Table Name	customers
Table Synonym	CTB
Table Comments	Stores personal and demographic information of all bank customers

Column Name	Data types	Description	Foreign Keys
customer_id	INT	Primary key - Unique customer identifier	PK
first_name	VARCHAR(50)	Customer's first name	
Last_name	VARCHAR(50)	Customer's last name	
middle_name	VARCHAR(50)	Customer's middle name	
date_of_birth	DATE	Customer's date of birth	
gender	ENUM('M','F','Other')	Customer's gender	
ssn	VARCHAR(20)	Social Security Number (unique)	
email	VARCHAR(100)	Customer's email address	
phone_number	VARCHAR(20)	Primary phone number	
alternate_phone	VARCHAR(20)	Secondary phone number	
address_line1	VARCHAR(100)	Primary address line 1	
address_line2	VARCHAR(100)	Primary address line 2	
city	VARCHAR(50)	City of residence	
state	VARCHAR(50)	State of residence	
zip_code	VARCHAR(10)	ZIP/Postal code	
country	VARCHAR(50)	Country of residence	
nationality	VARCHAR(50)	Customer's nationality	
occupation	VARCHAR(50)	Current occupation	
employer	VARCHAR(100)	Current employer name	
annual_income	DECIMAL(15,2)	Annual income in USD	
credit_score	INT	Credit score (300-850)	
customer_since	DATE	Date customer joined	

status	ENUM('Active','Inactive','Suspended','Closed')	Customer account status	
risk_level	ENUM('Low','Medium','High','Very High')	Risk assessment level	
is_verified	BOOLEAN	Identity verification status	
id_type	VARCHAR(30)	Identification document type	
id_number	VARCHAR(50)	Identification document number	
id_issue_date	DATE	ID document issue date	
id_expiry_date	DATE	ID document expiry date	
created_at	TIMESTAMP	Record creation timestamp	
updated_at	TIMESTAMP	Record last update timestamp	

Table Name	accounts
Table Synonym	ACT
Table Comments	Manages all bank accounts including checking, savings, and business accounts.

Column Name	Data Type	Description	Foreign Key
account_id	INT	Primary key - Unique account identifier	PK
customer_id	INT	Reference to account owner	FK → customers.customer_id
account_number	VARCHAR(20)	Unique account number	-

account_type	ENUM('Checking','Savings','Money Market','Certificate of Deposit','Business')	Type of account	-
account_subtype	VARCHAR(30)	Sub-category of account type	-
account_name	VARCHAR(100)	Custom account name	-
currency_code	VARCHAR(3)	Account currency (default USD)	-
balance	DECIMAL(15,2)	Current account balance	-
available_balance	DECIMAL(15,2)	Available funds for withdrawal	-
interest_rate	DECIMAL(5,2)	Annual interest rate	-
overdraft_limit	DECIMAL(15,2)	Maximum overdraft allowed	-
overdraft_protect	BOOLEAN	Overdraft protection enabled	-
min_balance_required	DECIMAL(15,2)	Minimum balance required	-
monthly_maintenance_fee	DECIMAL(10,2)	Monthly service fee	-
opened_date	DATE	Account opening date	-
closed_date	DATE	Account closing date (if applicable)	-
maturity_date	DATE	Maturity date for CDs	-

status	ENUM('Active','Dormant','Frozen','Closed')	Account status	-
is_joint	BOOLEAN	Joint account indicator	-
joint_owner1_id	INT	First joint owner reference	FK → customers.customer_id
joint_owner2_id	INT	Second joint owner reference	FK → customers.customer_id
joint_owner3_id	INT	Third joint owner reference	FK → customers.customer_id
atm_card_issued	BOOLEAN	ATM card issued indicator	-
atm_card_number	VARCHAR(20)	ATM card number	-
atm_pin_hash	VARCHAR(255)	Hashed ATM PIN	-
checkbook_issued	BOOLEAN	Checkbook issued indicator	-
last_transaction_date	DATE	Date of last transaction	-
statement_cycle	ENUM('Monthly','Quarterly','Annual')	Statement frequency	-
statement_method	ENUM('Mail','Email','Online Only')	Statement delivery method	-
notification_preferences	TEXT	User notification settings	-
created_at	TIMESTAMP	Record creation timestamp	-

updated_at	TIMESTAMP	Record last update timestamp	
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Table Name	Transactions
Table Synonym	TRAN
Table Comments	Records all financial transactions across accounts

Column Name	Data Type	Description	Foreign Key
transaction_id	INT	Primary key - Unique transaction identifier	PK
account_id	INT	Account involved in transaction	FK → accounts.account_id
transaction_reference	VARCHAR(50)	Unique transaction reference number	-
transaction_type	ENUM('Deposit', 'Withdrawal', 'Transfer', 'Payment', 'Fee', 'Interest', 'Refund', 'Chargeback')	Type of transaction	-
transaction_subtype	VARCHAR(30)	Sub-category of transaction	-
amount	DECIMAL(15,2)	Transaction amount	-
fee_amount	DECIMAL(10,2)	Fee charged for transaction	-
tax_amount	DECIMAL(10,2)	Tax applied to transaction	-
balance_before	DECIMAL(15,2)	Account balance before transaction	-
balance_after	DECIMAL(15,2)	Account balance after transaction	-
currency_code	VARCHAR(3)	Transaction currency	-

exchange_rate	DECIMAL(10,4)	Exchange rate applied (if foreign)	-
transaction_date	DATE TIME	Date and time of transaction	-
value_date	DATE	Value date of transaction	-
posting_date	DATE	Date transaction was posted	-
description	VARCHAR(255))	Transaction description	-
memo	VARCHAR(255))	Internal memo/notes	-
category	VARCHAR(50)	Transaction category	-
subcategory	VARCHAR(50)	Transaction subcategory	-
merchant_name	VARCHAR(100))	Merchant name (if applicable)	-
merchant_category_code	VARCHAR(10)	MCC code for merchant	-
terminal_id	VARCHAR(30)	Terminal/device identifier	-
authorization_code	VARCHAR(50)	Transaction authorization code	-
reference_number	VARCHAR(50)	Reference number from partner	-
trace_number	VARCHAR(50)	Trace number for tracking	-
status	ENUM('Pending', 'Completed','Failed', 'Reversed','Cancelled')	Transaction status	-
is_international	BOOLEAN	International transaction flag	-
is_online	BOOLEAN	Online transaction flag	-
source_channel	ENUM('Branch','ATM','Mobile','Online Banking','Call Center','POS','Wire Transfer')	Transaction channel	-

ip_address	VARCHAR(45)	IP address of transaction origin	-
device_id	VARCHAR(100))	Device identifier	-
location	VARCHAR(100))	Transaction location	-
fraud_check_flag	BOOLEAN	Fraud detection flag	-
approval_code	VARCHAR(20)	Approval code from processor	-
settlement_date	DATE	Settlement date for transaction	

Table Name	loans
Table Synonym	LON
Table Comments	Manages all loan products including personal, auto, mortgage, and business loans.

Column Name	Data Type	Description	Foreign Key
loan_id	INT	Primary key - Unique loan identifier	PK
customer_id	INT	Loan borrower reference	FK → customers.customer_id
account_id	INT	Associated account for payments	FK → accounts.account_id
loan_number	VARCHAR(20)	Unique loan number	-
loan_type	ENUM('Personal', 'Auto', 'Mortgage', 'Home Equity', 'Student', 'Business', 'Payday'))	Type of loan product	-

loan_subtype	VARCHAR(30)	Sub-category of loan	-
principal_amount	DECIMAL(15,2)	Original loan amount	-
interest_rate	DECIMAL(6,2)	Annual interest rate	-
apr	DECIMAL(6,2)	Annual Percentage Rate	-
loan_term_months	INT	Loan term in months	-
monthly_payment	DECIMAL(10,2)	Required monthly payment	-
origination_fee	DECIMAL(10,2)	Fee for loan origination	-
processing_fee	DECIMAL(10,2)	Loan processing fee	-
late_fee	DECIMAL(10,2)	Late payment penalty	-
prepayment_penalty	DECIMAL(10,2)	Penalty for early repayment	-
current_balance	DECIMAL(15,2)	Remaining loan balance	-
outstanding_principal	DECIMAL(15,2)	Principal amount remaining	-
accrued_interest	DECIMAL(15,2)	Interest accrued to date	-
next_payment_date	DATE	Date of next payment due	-
last_payment_date	DATE	Date of last payment	-
last_payment_amount	DECIMAL(10,2)	Amount of last payment	-
collateral_type	VARCHAR(50)	Type of collateral (if secured)	-
collateral_value	DECIMAL(15,2)	Value of collateral	-
co_signer_id	INT	Co-signer reference	FK → customers.customer_id
approval_date	DATE	Date loan was approved	-
disbursement_date	DATE	Date funds were disbursed	-
maturity_date	DATE	Loan maturity/end date	-
status	ENUM('Pending','Approved','Active','Delinquent','Defaulted','Paid Off','Charged Off')	Loan status	-

risk_rating	ENUM('A','B','C','D','E')	Risk assessment rating	-
purpose	VARCHAR(255)	Purpose of loan	-
employment_verification	BOOLEAN	Employment verified flag	-
income_verification	BOOLEAN	Income verified flag	-
credit_approval_level	ENUM('Auto','Manual','Supervisory')	Approval workflow level	-
notes	TEXT	Additional notes/comments	-
created_at	TIMESTAMP	Record creation timestamp	-
updated_at	TIMESTAMP	Record last update timestamp	-

Table Name	Loan_payments
Table Synonym	LONP
Table Comments	Tracks all payments made towards loans

Column Name	Data Type	Description	Foreign Key
payment_id	INT	Primary key - Unique payment identifier	PK
loan_id	INT	Loan being paid	FK → loans.loan_id
payment_reference	VARCHAR(50)	Unique payment reference number	-
payment_date	DATE	Date of payment	-
amount	DECIMAL(15,2)	Total payment amount	-
principal_paid	DECIMAL(15,2)	Portion applied to principal	-
interest_paid	DECIMAL(15,2)	Portion applied to interest	-
fee_paid	DECIMAL(15,2)	Fees paid	-

late_charge_paid	DECIMAL(15,2)	Late fees paid	-
payment_method	ENUM('Bank Transfer','Check','Cash','Debit Card','Auto Debit')	Method of payment	-
payment_status	ENUM('Pending','Completed','Failed','Reversed')	Payment status	-
transaction_id	INT	Associated transaction	FK → transactions.transaction_id
remaining_balance	DECIMAL(15,2)	Balance after payment	-
days_late	INT	Days payment was late	-
is_auto_payment	BOOLEAN	Auto-debit payment flag	-
auto_payment_id	INT	Auto-payment setup ID	-
receipt_number	VARCHAR(50)	Payment receipt number	-
notes	TEXT	Additional payment notes	-
created_at	TIMESTAMP	Record creation timestamp	

Table Name	Credit_cards
Table Synonym	CRC
Table Comments	Manages credit card accounts and their details

Column Name	Data Type	Description	Foreign Key
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card_id	INT	Primary key - Unique card identifier	PK
customer_id	INT	Cardholder reference	FK → customers.customer_id
account_id	INT	Associated account for billing	FK → accounts.account_id
card_number	VARCHAR(20)	Primary Account Number (PAN)	-
card_type	ENUM('Visa','MasterCard','American Express','Discover')	Credit card network	-
card_level	ENUM('Standard','Gold','Platinum','Signature','Black')	Card tier/level	-
card_holder_name	VARCHAR(100)	Name on card	-
expiry_date	DATE	Card expiration date	-
cvv_hash	VARCHAR(255)	Hashed CVV code	-
credit_limit	DECIMAL(15,2)	Maximum credit limit	-
available_credit	DECIMAL(15,2)	Remaining available credit	-
current_balance	DECIMAL(15,2)	Current outstanding balance	-
minimum_payment	DECIMAL(10,2)	Minimum monthly payment	-
apr_purchase	DECIMAL(6,2)	APR for purchases	-
apr_cash_advance	DECIMAL(6,2)	APR for cash advances	-
apr_balance_transfer	DECIMAL(6,2)	APR for balance transfers	-

annual_fee	DECIMAL(10,2)	Annual card fee	-
foreign_transaction_fee	DECIMAL(5,2)	Fee for foreign transactions	-
late_payment_fee	DECIMAL(10,2)	Late payment penalty	-
cash_advance_fee	DECIMAL(5,2)	Cash advance fee	-
reward_points_balance	INT	Accumulated reward points	-
reward_program	VARCHAR(50)	Rewards program name	-
cashback_percentage	DECIMAL(5,2)	Cashback rate	-
statement_cycle_day	INT	Day of month for statement	-
payment_due_days	INT	Days after statement for payment	-
status	ENUM('Active','Suspended','Cancelled','Expired')	Card status	-
issued_date	DATE	Date card was issued	-
cancelled_date	DATE	Date card was cancelled	-
last_activity_date	DATE	Last card usage date	-
credit_score_at_issue	INT	Credit score at issuance	-
is_chip_enabled	BOOLEAN	Chip technology flag	-
is_contactless	BOOLEAN	Contactless payment flag	-
pin_set	BOOLEAN	PIN has been set	-
fraud_alerts_enabled	BOOLEAN	Fraud alerts active	-
monthly_statement_enabled	BOOLEAN	Statement generation active	

Table Name	Card_transactions
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Table Synonym	CTR
Table Comments	Details of transactions made using credit cards

Column Name	Data Type	Description	Foreign Key
card_transaction_id	INT	Primary key - Unique identifier	PK
card_id	INT	Card used for transaction	FK → credit_cards.card_id
transaction_id	INT	Associated transaction	FK → transactions.transaction_id
merchant_name	VARCHAR(100)	Merchant name	-
merchant_id	VARCHAR(30)	Merchant identifier	-
merchant_city	VARCHAR(50)	Merchant city	-
merchant_state	VARCHAR(50)	Merchant state	-
merchant_country	VARCHAR(50)	Merchant country	-
merchant_category_code	VARCHAR(10)	MCC code	-
transaction_type	ENUM('Purchase','Cash Advance','Balance Transfer','Refund','Fee')	Type of card transaction	-
purchase_amount	DECIMAL(15,2)	Amount of purchase	-

cashback_amount	DECIMAL(10,2)	Cashback earned	-
reward_points_earned	INT	Points earned from transaction	-
authorization_code	VARCHAR(50)	Transaction authorization code	-
pos_entry_mode	VARCHAR(20)	POS entry method	-
card_present	BOOLEAN	Physical card present	-
chip_used	BOOLEAN	EMV chip used	-
pin_verified	BOOLEAN	PIN verified	-
signature_required	BOOLEAN	Signature required	

Table Name	Investment_accounts
Table Synonym	IVA
Table Comments	Manages investment accounts including brokerage, IRA, and retirement accounts

Column Name	Data Type	Description	Foreign Key
investment_id	INT	Primary key - Unique investment identifier	PK
customer_id	INT	Account holder reference	FK → customers.customer_id

account_id	INT	Associated bank account	FK → accounts.account_id
account_number	VARCHAR(20)	Unique investment account number	-
account_type	ENUM('Brokerage','IRA Traditional','IRA Roth','401(k)','Pension','529 Plan','Trust','Mutual Fund','Annuity')	Type of investment account	-
investment_advisor	VARCHAR(100)	Advisor name	-
advisor_contact	VARCHAR(50)	Advisor contact information	-
risk_tolerance	ENUM('Conservative','Moderate','Aggressive','Speculative')	Client's risk tolerance	-
investment_objective	VARCHAR(100)	Investment goals	-
time_horizon	VARCHAR(50)	Investment time frame	-
total_value	DECIMAL(15,2)	Total portfolio value	-
total_cost_basis	DECIMAL(15,2)	Total cost of investments	-
total_gain_loss	DECIMAL(15,2)	Total unrealized gain/loss	-
cash_balance	DECIMAL(15,2)	Cash held in account	-
margin_balance	DECIMAL(15,2)	Margin loan balance	-
margin_interest_rate	DECIMAL(6,2)	Margin loan interest rate	-
annual_return_percentage	DECIMAL(5,2)	Annual return rate	-

last_performance_date	DATE	Last performance update	-
opened_date	DATE	Account opening date	-
closed_date	DATE	Account closing date	-
status	ENUM('Active','Dormant','Closed','Suspended')	Account status	-
is_taxable	BOOLEAN	Taxable account flag	-
tax_bracket	DECIMAL(5,2)	Tax bracket percentage	-
beneficiary_name	VARCHAR(100)	Beneficiary name	-
beneficiary_relationship	VARCHAR(50)	Relationship to beneficiary	-
beneficiary_ssn	VARCHAR(20)	Beneficiary SSN	-
trust_name	VARCHAR(100)	Trust name (if applicable)	-
trust_contact	VARCHAR(100)	Trust contact info	-

Table Name	Securities
Table Synonym	IVA
Table Comments	Master list of securities available for investment

Column Name	Data Type	Description	Foreign Key
security_id	INT	Primary key - Unique security identifier	PK
ticker_symbol	VARCHAR(10)	Stock ticker symbol	-
security_name	VARCHAR(100)	Full security name	-
security_type	ENUM('Stock','Bond','ETF','Mutual Fund','Option','Future','Forex','Commodity')	Type of security	-

sector	VARCHAR(50)	Industry sector	-
industry	VARCHAR(50)	Specific industry	-
country	VARCHAR(50)	Country of origin	-
currency_code	VARCHAR(3)	Trading currency	-
current_price	DECIMAL(15,4)	Current market price	-
previous_price	DECIMAL(15,4)	Previous close price	-
day_change	DECIMAL(10,4)	Price change from previous	-
day_change_percentage	DECIMAL(6,2)	Percentage price change	-
volume	INT	Current trading volume	-
avg_volume	INT	Average daily volume	-
market_cap	DECIMAL(20,2)	Market capitalization	-
pe_ratio	DECIMAL(10,2)	Price-to-earnings ratio	-
dividend_yield	DECIMAL(6,2)	Dividend yield percentage	-
dividend_amount	DECIMAL(10,4)	Dividend per share	-
ex_dividend_date	DATE	Ex-dividend date	-
earnings_per_share	DECIMAL(10,2)	EPS for fiscal year	-
beta	DECIMAL(6,2)	Volatility measure	-
rating	VARCHAR(10)	Analyst rating	-
target_price	DECIMAL(15,4)	Analyst target price	-
issued_date	DATE	Issue date	-
maturity_date	DATE	Maturity date (for bonds)	-
coupon_rate	DECIMAL(6,2)	Bond coupon rate	-
bond_rating	VARCHAR(10)	Bond credit rating	-
is_active	BOOLEAN	Security active status	-
created_at	TIMESTAMP	Record creation timestamp	-

updated_at	TIMESTAMP	Record last update timestamp	
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Table Name	holdings
Table Synonym	HLD
Table Comments	Tracks securities held in investment accounts

Column Name	Data Type	Description	Foreign Key
holding_id	INT	Primary key - Unique holding identifier	PK
investment_id	INT	Investment account reference	FK → investment_accounts.investment_id
security_id	INT	Security held	FK → securities.security_id
quantity	DECIMAL(15,4)	Number of shares/units held	-
average_cost	DECIMAL(15,4)	Average cost basis per share	-
current_value	DECIMAL(15,4)	Current market value	-
cost_basis	DECIMAL(15,4)	Total cost basis	-
gain_loss	DECIMAL(15,4)	Unrealized gain/loss	-
gain_loss_percent age	DECIMAL(6,2)	Gain/loss percentage	-

purchase_date	DATE	Date of purchase	-
purchase_price	DECIMAL(15,4))	Purchase price per share	-
last_sale_date	DATE	Last sale date	-
last_sale_price	DECIMAL(15,4))	Last sale price	-
dividend_reinvest	BOOLEAN	Dividend reinvestment flag	-
tax_lot_number	VARCHAR(30)	Tax lot identifier	-
holding_type	ENUM('Long','Short')	Long or short position	-
status	ENUM('Active','Sold','Partially Sold')	Holding status	-
notes	TEXT	Additional notes	

Table Name	branches
Table Synonym	BRC
Table Comments	Contains information about all branch locations

Column Name	Data Type	Description	Foreign Key
branch_id	INT	Primary key - Unique branch identifier	PK
branch_code	VARCHAR(10)	Unique branch code	-
branch_name	VARCHAR(100)	Full branch name	-
branch_type	ENUM('Head Office','Regional','Local','Mobile Unit','Virtual')	Type of branch	-

address_line1	VARCHAR(100)	Branch address line 1	-
address_line2	VARCHAR(100)	Branch address line 2	-
city	VARCHAR(50)	Branch city	-
state	VARCHAR(50)	Branch state	-
zip_code	VARCHAR(10)	Branch ZIP code	-
country	VARCHAR(50)	Branch country	-
phone_number	VARCHAR(20)	Branch phone number	-
fax_number	VARCHAR(20)	Branch fax number	-
email	VARCHAR(100)	Branch email address	-
manager_name	VARCHAR(100)	Branch manager name	-
manager_id	INT	Branch manager employee ID	-
region	VARCHAR(50)	Geographic region	-
timezone	VARCHAR(50)	Time zone	-
operating_hours	VARCHAR(100)	Regular operating hours	-
weekend_hours	VARCHAR(100)	Weekend operating hours	-
holiday_hours	TEXT	Holiday hours	-
establishment_date	DATE	Branch opening date	-
square_footage	INT	Building size in sq ft	-
number_of_atms	INT	Number of ATMs at branch	-
number_of_tellers	INT	Number of teller stations	-
has_safe_deposit	BOOLEAN	Safe deposit boxes available	-
has_drive_thru	BOOLEAN	Drive-through service available	-
has_parking	BOOLEAN	Parking available	-
has_wheelchair_access	BOOLEAN	Wheelchair accessible	-

is_active	BOOLEAN	Branch active status	-
notes	TEXT	Additional branch notes	-
created_at	TIMESTAMP	Record creation timestamp	-
updated_at	TIMESTAMP	Record last update timestamp	-

Table Name	employess
Table Synonym	EMP
Table Comments	Manages employee records and information

Column Name	Data Type	Description	Foreign Key
employee_id	INT	Primary key - Unique employee identifier	PK
branch_id	INT	Assigned branch reference	FK → branches.branch_id
employee_number	VARCHAR(20)	Unique employee number	-
first_name	VARCHAR(50)	Employee first name	-
last_name	VARCHAR(50)	Employee last name	-
middle_name	VARCHAR(50)	Employee middle name	-
date_of_birth	DATE	Employee date of birth	-
ssn	VARCHAR(20)	Employee SSN (unique)	-
email	VARCHAR(100)	Employee email	-
phone_number	VARCHAR(20)	Employee phone	-
address_line1	VARCHAR(100)	Employee address line 1	-

address_line2	VARCHAR(100)	Employee address line 2	-
city	VARCHAR(50)	Employee city	-
state	VARCHAR(50)	Employee state	-
zip_code	VARCHAR(10)	Employee ZIP code	-
country	VARCHAR(50)	Employee country	-
position	VARCHAR(50)	Job position title	-
department	VARCHAR(50)	Department name	-
title	VARCHAR(50)	Job title	-
role	ENUM('Teller','Customer Service','Loan Officer','Branch Manager','Regional Manager','Investment Advisor','Compliance Officer','Fraud Analyst','IT Staff','Executive')	Role category	-
supervisor_id	INT	Supervisor reference	FK → employees.employee_id
hire_date	DATE	Date of hire	-
termination_date	DATE	Date of termination	-
salary	DECIMAL(15,2)	Annual salary	-

bonus_percentage	DECIMAL(5,2)	Bonus percentage	-
benefits_package	VARCHAR(50)	Benefits package code	-
vacation_days_accrued	INT	Vacation days earned	-
sick_days_accrued	INT	Sick days earned	-
performance_rating	ENUM('Excellent','Good','Satisfactory','Needs Improvement','Unsatisfactory')	Performance rating	-
last_performance_review	DATE	Date of last review	-
training_completed	TEXT	Completed training list	-
certifications	TEXT	Professional certifications	-
security_clearance_level	INT	Security clearance level (1-5)	-
is_active	BOOLEAN	Employee active status	-
emergency_contact_name	VARCHAR(100)	Emergency contact name	-
emergency_contact_phone	VARCHAR(20)	Emergency contact phone	-
created_at	TIMESTAMP	Record creation timestamp	-
updated_at	TIMESTAMP	Record last update timestamp	-

Table Name	Teller_transactions
Table Synonym	TTA
Table Comments	Tracks teller-related transaction activities.

Column Name	Data Type	Description	Foreign Key
teller_transaction_id	INT	Primary key - Unique identifier	PK
transaction_id	INT	Associated transaction	FK → transactions.transaction_id
employee_id	INT	Employee (teller) reference	FK → employees.employee_id
branch_id	INT	Branch reference	FK → branches.branch_id
teller_number	VARCHAR(10)	Teller station number	-
start_time	DATETIME	Transaction start time	-
end_time	DATETIME	Transaction end time	-
cash_balance_start	DECIMAL(15,2)	Cash balance at start	-
cash_balance_end	DECIMAL(15,2)	Cash balance at end	-
checks_processed	INT	Number of checks processed	-
check_amount_total	DECIMAL(15,2)	Total check amount	-
cash_deposits	INT	Number of cash deposits	-
cash_deposit_total	DECIMAL(15,2)	Total cash deposits	-
cash_withdrawals	INT	Number of cash withdrawals	-
cash_withdrawal_total	DECIMAL(15,2)	Total cash withdrawals	-

transfers_processed	INT	Number of transfers	-
transfer_total	DECIMAL(15,2)	Total transfer amount	-

Table Name	Online_banking_users
Table Synonym	OBU
Table Comments	Manages online and mobile banking user credentials

Column Name	Data Type	Description	Foreign Key
online_user_id	INT	Primary key - Unique user identifier	PK
customer_id	INT	Customer reference	FK → customers.customer_id
username	VARCHAR(50)	Login username	-
password_hash	VARCHAR(255)	Hashed password	-
salt	VARCHAR(100)	Password salt value	-
security_question1	VARCHAR(100)	Security question 1	-
security_answer1	VARCHAR(100)	Answer to question 1	-
security_question2	VARCHAR(100)	Security question 2	-
security_answer2	VARCHAR(100)	Answer to question 2	-

security_question3	VARCHAR(100)	Security question 3	-
security_answer3	VARCHAR(100)	Answer to question 3	-
two_factor_enabled	BOOLEAN	2FA enabled flag	-
two_factor_phone	VARCHAR(20)	Phone for 2FA	-
two_factor_email	VARCHAR(100)	Email for 2FA	-
last_login	DATE TIME	Last login timestamp	-
last_login_ip	VARCHAR(45)	IP of last login	-
failed_login_attempts	INT	Count of failed attempts	-
account_locked_until	DATE TIME	Lock expiry time	-
is_locked	BOOLEAN	Account locked flag	-
is_active	BOOLEAN	User active status	-
last_password_change	DATE TIME	Last password change date	-
password_expiry	DATE TIME	Password expiry date	-
login_notifications_enabled	BOOLEAN	Login alerts enabled	-
transaction_notifications_enabled	BOOLEAN	Transaction alerts enabled	-
alert_preferences	TEXT	Alert preferences JSON	-
mobile_app_registered	BOOLEAN	Mobile app registered	-
mobile_device_id	VARCHAR(100)	Mobile device ID	-
push_token	VARCHAR(255)	Push notification token	-
created_at	TIME STAMP	Record creation timestamp	-

updated_at	TIMESTAMP	Record last update timestamp	-
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Table Name	Atm_machines
Table Synonym	ATM
Table Comments	Manages ATM machine inventory and status

Column Name	Data Type	Description	Foreign Key
atm_id	INT	Primary key - Unique ATM identifier	PK
branch_id	INT	Associated branch	FK → branches.branch_id
atm_code	VARCHAR(20)	Unique ATM code	-
atm_location	VARCHAR(100)	ATM location description	-
address_line1	VARCHAR(100)	ATM address line 1	-
address_line2	VARCHAR(100)	ATM address line 2	-
city	VARCHAR(50)	ATM city	-
state	VARCHAR(50)	ATM state	-
zip_code	VARCHAR(10)	ATM ZIP code	-
latitude	DECIMAL(10,8)	Geographic latitude	-
longitude	DECIMAL(11,8)	Geographic longitude	-
atm_type	ENUM('On-Site','Off-Site','Mobile')	ATM location type	-
manufacturer	VARCHAR(50)	ATM manufacturer	-
model	VARCHAR(50)	ATM model number	-

serial_number	VARCHAR(50)	ATM serial number	-
software_version	VARCHAR(30)	ATM software version	-
cash_capacity	DECIMAL(15,2)	Maximum cash capacity	-
current_cash_level	DECIMAL(15,2)	Current cash on hand	-
currency_denominations	TEXT	Available denominations	-
transaction_count_today	INT	Today's transaction count	-
transaction_count_monthly	INT	Monthly transaction count	-
last_maintenance_date	DATE	Last maintenance date	-
next_maintenance_date	DATE	Next scheduled maintenance	-
status	ENUM('Online','Offline','Out of Service','Under Maintenance')	ATM status	-
has_deposit_capability	BOOLEAN	Deposit capability	-
has_check_capability	BOOLEAN	Check deposit capability	-
has_envelope_free	BOOLEAN	Envelope-free deposit	-
has_cardless_capability	BOOLEAN	Cardless access	-
has_nfc_capability	BOOLEAN	NFC capability	-
screen_size	VARCHAR(20)	Screen size description	-
accessibility_features	TEXT	Accessibility features	-

installed_date	DATE	Installation date	-
notes	TEXT	Additional notes	-
created_at	TIMESTAMP	Record creation timestamp	-
updated_at	TIMESTAMP	Record last update timestamp	-

Table Name	Atm_transactions
Table Synonym	ATT
Table Comments	Records ATM transaction details and processing

Column Name	Data Type	Description	Foreign Key
atm_transaction_id	INT	Primary key - Unique identifier	PK
transaction_id	INT	Associated transaction	FK → transactions.transaction_id
atm_id	INT	ATM machine reference	FK → atm_machines.atm_id
customer_id	INT	Customer performing transaction	FK → customers.customer_id
atm_card_number	VARCHAR(20)	Card number used	-
card_present	BOOLEAN	Physical card present	-

entry_mode	VARCHAR(20)	Card entry method	-
terminal_id	VARCHAR(30)	Terminal identifier	-
transaction_sequence_number	INT	Sequence number	-
response_code	VARCHAR(10)	Response code from ATM	-
response_message	VARCHAR(255)	Response message	-
processing_time	DECIMAL(5,2)	Processing time in seconds	-
start_time	DATE TIME	Transaction start time	-
completion_time	DATE TIME	Transaction completion time	-
error_code	VARCHAR(20)	Error code if any	-
error_description	TEXT	Error description	

Table Name	Merchant_accounts
Table Synonym	MRA
Table Comments	Manages merchant accounts for payment processing

Column Name	Data Type	Description	Foreign Key
merchant_id	INT	Primary key - Unique merchant identifier	PK
merchant_name	VARCHAR(100)	Merchant business name	-
merchant_category	VARCHAR(50)	Business category	-
merchant_subcategory	VARCHAR(50)	Business subcategory	-
business_type	VARCHAR(50)	Type of business	-
tax_id	VARCHAR(30)	Tax ID (EIN)	-
dba_name	VARCHAR(100)	Doing Business As name	-
address_line1	VARCHAR(100)	Address line 1	-
address_line2	VARCHAR(100)	Address line 2	-
city	VARCHAR(50)	City	-
state	VARCHAR(50)	State	-
zip_code	VARCHAR(10)	ZIP code	-
country	VARCHAR(50)	Country	-
phone_number	VARCHAR(20)	Contact phone	-
email	VARCHAR(100)	Contact email	-
website	VARCHAR(100)	Business website	-
contact_person	VARCHAR(100)	Primary contact name	-
contact_phone	VARCHAR(20)	Contact phone	-
contact_email	VARCHAR(100)	Contact email	-
account_number	VARCHAR(20)	Merchant account number	-
settlement_account_id	INT	Settlement account reference	-
processing_fee_rate	DECIMAL(6,2)	Processing fee percentage	-
discount_rate	DECIMAL(6,2)	Discount rate	-
transaction_fee	DECIMAL(5,2)	Per transaction fee	-
monthly_fee	DECIMAL(10,2)	Monthly service fee	-

chargeback_fee	DECIMAL(10,2)	Chargeback fee	-
reserve_percentage	DECIMAL(6,2)	Reserve hold percentage	-
reserve_amount	DECIMAL(15,2)	Reserve hold amount	-
average_ticket_size	DECIMAL(10,2)	Average transaction size	-
monthly_volume	DECIMAL(15,2)	Monthly processing volume	-
annual_volume	DECIMAL(15,2)	Annual processing volume	-
risk_rating	ENUM('Low','Medium','High','Very High')	Risk assessment	-
status	ENUM('Active','Suspended','Terminated','Pending')	Account status	-
approved_date	DATE	Approval date	-
terminated_date	DATE	Termination date	-
business_start_date	DATE	Business start date	-
employee_count	INT	Number of employees	-
annual_revenue	DECIMAL(15,2)	Annual revenue	-
notes	TEXT	Additional notes	-
created_at	TIMESTAMP	Record creation timestamp	-
updated_at	TIMESTAMP	Record last update timestamp	-

Table Name	bill_payments
Table Synonym	BLP
Table Comments	Manages bill payments made by customers

Column Name	Data Type	Description	Foreign Key
bill_payment_id	INT	Primary key - Unique identifier	PK

customer_id	INT	Customer making payment	FK → customers.customer_id
account_id	INT	Account used for payment	FK → accounts.account_id
transaction_id	INT	Associated transaction	FK → transactions.transaction_id
payee_name	VARCHAR(100)	Payee/recipient name	-
payee_account_number	VARCHAR(30)	Payee account number	-
payee_routing_number	VARCHAR(20)	Payee routing number	-
payee_address	VARCHAR(255)	Payee address	-
payee_city	VARCHAR(50)	Payee city	-
payee_state	VARCHAR(50)	Payee state	-
payee_zip	VARCHAR(10)	Payee ZIP code	-
bill_category	ENUM('Utilities','Insurance','Loan','Credit Card','Rent','Mortgage','Subscription','Tax','Other')	Category of bill	-
bill_subcategory	VARCHAR(50)	Subcategory of bill	-
amount	DECIMAL(15,2)	Payment amount	-
due_date	DATE	Bill due date	-
payment_date	DATE	Date payment was made	-

confirmation_number	VARCHAR(50)	Payment confirmation number	-
processing_fee	DECIMAL(10,2)	Processing fee	-
schedule_type	ENUM('One-Time','Recurring','Auto-Pay')	Payment schedule type	-
recurrence_frequency	VARCHAR(20)	Frequency of recurrence	-
next_scheduled_date	DATE	Next scheduled payment date	-
is_recurring	BOOLEAN	Recurring payment flag	-
is_auto_pay	BOOLEAN	Auto-pay enabled flag	-
payment_statuses	ENUM('Scheduled','Pending','Processing','Completed','Failed','Cancelled')	Payment status	-
failure_reason	VARCHAR(255)	Reason for failure if any	-
notes	TEXT	Additional notes	-
created_at	TIMESTAMP	Record creation timestamp	-
updated_at	TIMESTAMP	Record last update timestamp	

Table Name	audit_logs
Table Synonym	ADL
Table Comments	Tracks all system changes for compliance and security

Column Name	Data Type	Description	Foreign Key
audit_id	INT	Primary key - Unique audit identifier	PK
table_name	VARCHAR(50)	Table that was changed	-

record_id	INT	ID of the record changed	-
action	ENUM('INSERT','UPDATE','DELETE','SELECT','LOGIN','LOGOUT','FAILED_LOGIN')	Action performed	-
field_name	VARCHAR(50)	Field that was changed	-
old_value	TEXT	Previous value	-
new_value	TEXT	New value	-
user_id	INT	User who performed action	-
user_type	ENUM('Customer','Employee','System')	Type of user	-
ip_address	VARCHAR(45)	IP address of user	-
user_agent	VARCHAR(255)	Browser/device info	-
session_id	VARCHAR(100)	Session identifier	-
timestamp	DATE TIME	Time of action	-
affected_rows	INT	Number of rows affected	-
is_sensitive	BOOLEAN	Sensitive action flag	-
compliance_category	VARCHAR(50)	Compliance category	-
audit_notes	TEXT	Additional audit notes	

Table Name	notifications
Table Synonym	NOT

Table Comments	Manages system notifications and alerts for customers and employees.
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Column Name	Data Type	Description	Foreign Key
notification_id	INT	Primary key - Unique notification identifier	PK
customer_id	INT	Customer recipient	FK → customers.customer_id
employee_id	INT	Employee recipient	FK → employees.employee_id
branch_id	INT	Branch recipient	FK → branches.branch_id
notification_type	ENUM('Alert','Warning','Information','Promotion','Reminder','Transaction Alert','Security Alert','Service Update')	Type of notification	-
priority	ENUM('Low','Medium','High','Critical')	Priority level	-
subject	VARCHAR(255)	Notification subject	-
message	TEXT	Notification message	-
link_url	VARCHAR(255)	Link to related content	-
read_status	BOOLEAN	Read/unread status	-
read_timestamp	DATETIME	Time notification was read	-
dismissed	BOOLEAN	Dismissed flag	-
dismissed_timestamp	DATETIME	Time notification was dismissed	-

sent_via_email	BOOLEAN	Sent via email	-
sent_via_sms	BOOLEAN	Sent via SMS	-
sent_via_push	BOOLEAN	Sent via push notification	-
sent_via_app	BOOLEAN	Sent via app	-
email_sent_status	ENUM('Pending','Sent','Failed','Bounced')	Email delivery status	-
sms_sent_status	ENUM('Pending','Sent','Failed')	SMS delivery status	-
push_sent_status	ENUM('Pending','Sent','Failed')	Push delivery status	-
scheduled_time	DATE TIME	Scheduled delivery time	-
sent_time	DATE TIME	Actual sent time	-
expiry_date	DATE TIME	Notification expiry date	-
created_by	INT	User who created notification	-
creation_type	ENUM('System','Manual','Triggered')	How notification was created	-
trigger_event	VARCHAR(50)	Event that triggered notification	-
related_transaction_id	INT	Related transaction	FK → transactions.transaction_id
related_account_id	INT	Related account	FK → accounts.account_id
related_loan_id	INT	Related loan	FK → loans.loan_id
additional_data	JSON	Additional data in JSON format	-

created_at	TIMESTAMP	Record creation timestamp	-
updated_at	TIMESTAMP	Record last update timestamp	